

1.2 Characteristics of Model Checking

This book is devoted to the principles of model checking:

Model checking is an automated technique that, given a finite-state model of a system and a formal property, systematically checks whether this property holds for (a given state in) that model.

The next chapters treat the elementary technical details of model checking. This section describes the process of model checking (how to use it), presents its main advantages and drawbacks, and discusses its role in the system development cycle.

1.2.1 The Model-Checking Process

In applying model checking to a design the following different phases can be distinguished:

- *Modeling* phase:
 - model the system under consideration using the model description language of the model checker at hand;
 - as a first sanity check and quick assessment of the model perform some simulations;
 - formalize the property to be checked using the property specification language.
- *Running* phase: run the model checker to check the validity of the property in the system model.
- *Analysis* phase:
 - property satisfied? → check next property (if any);
 - property violated? →
 1. analyze generated counterexample by simulation;
 2. refine the model, design, or property;
 3. repeat the entire procedure.
 - out of memory? → try to reduce the model and try again.